Project 2 proposal for team Tigers

Datasci 200 - Kleeman, Fall 2026

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GitHub repository (name and link): https://github.com/justinye617/200-FA2026-EDA-Tigers

Planned mode of communication: Primary mode will be Slack, followed by email, and iMessage. We expect to meet approximately once per week Mon-Th at 9pm PT. Additional meetings will be added in these time windows as the project deadline becomes closer in weeks 13 and 14.

Primary Dataset and Description:
https://huggingface.co/datasets/frankjc2022/steam-dataset/viewer/games/train?p=2
Our data set is a public one that contains information for video games on Steam, a digital storefront primarily used for PC games. The dataset is approximately 48K rows and represents a sampling of games with various release dates. It is not an exhaustive list of games on Steam.

Within the dataset, the most interesting columns include information about each game entry. Specifically, prices, platforms, genres, number of user recommendations, metacritic scores, publishers, and release dates are among the notably useful elements. Given that we know the dataset is not exhaustive, nor is it necessarily a representative sample of all games on Steam, we will not seek to make conclusions about the broader market or release volumes. Since it is public, it has likely been appended several times and will have some holes.

However, we hope to learn if there are patterns of note for the games in the dataset between metacritic scores (the rating given to games by an average of critics across global media publishers), and the type of game genre, or platform on which it is available. We will also explore whether critic ratings necessarily correlate with user recommendations or whether certain publishers seem to focus on specific release dates during the year. Questions such as, “are better rated games more often found in a certain game genre within this data set?” or “are the games with the largest number of user recommendations more often free to play?” are ones we expect to tackle.

List of columns that are most interesting and what we hope to learn from them:

- **Metacritic score and User Recommendations:** What signal does this provide when it comes to the game genre, cost, or publisher?
- **Release dates:** Are these somehow more correlated with game success? Are certain publishers more likely to release games in specific seasonal windows?
- **Genre:** Are certain genres more popular? Do they command a higher price point from players? Has this changed over time?
- **Publishers:** For those with multiple releases, to what degree does concentrating in a genre vs releasing games across various genres lead to a larger volume of user recommendations?
- **Price**: Are free to play games better rated than those for which players have to pay up front?

Generally, we will look to see what story emerges from player and critic ratings of games (a proxy for game quality) in light of the game genre, price, and release timing.

Supplemental data sets we might use to enrich the analysis:

Given that we have app IDs and the name of certain Steam games in this dataset, we may be able to enrich the analysis with other data sets that share these fields and can be joined. Further metadata such as concurrent users on Steam (since the purchase portal operates as a live service) or detailed information from user reviews will be useful to get an estimate of the game's relative performance with players. In the same file folder as the original dataset, there exist separate files that are organized by the same app ID with some of this information, so we will look to combine the data through joins that make sense to further our understanding.

Timeline of milestones from now through final report submission:

- **By Monday, November 9th -**
    - Have cleaned and prepared the data set for analysis
    - Pull in any supplemental data sets and join them
    - Begin writeup of data preparation and cleaning for the final paper
    - Discuss which specific sections will be part of the EDA and findings portion of the paper. Divide authorship responsibility for those sections up among team members (target of ~300 words each)
- **By Monday, November 16th -**
    - Have started analysis on EDA sections (both graphical and narrative writeups)
    - Deadline for completion of Homework 10-11
- **By Monday, November 23rd** (week of Thanksgiving)
    - Complete writeup of Introduction and Guiding questions sections
    - Complete writeup of Assumptions and Limitations section
    - Complete most of EDA analysis (code complete)
    - Have at least started Homework 12-13
    - Begin authorship of Powerpoint presentation using graphical results
- **By Monday, November 30th** -
    - Complete EDA analysis narrative writeup
    - Complete Powerpoint presentation
- **By Monday, December 7th** - Submit Homework 12-13
- **By Tuesday, December 8th** - Final project submission
    - Complete writeup of Conclusion
    - Do final editing to ensure story/flow
    - Author final titles and flow in the Powerpoint presentation